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Trade and life expectancy in China: a cointegration analysis

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ABSTRACT
The purpose behind this paper is to empirically examine the relationship between trade openness and life expectancy for the Chinese economy. Data are collected for the period 1970–2015. Autoregressive Distributed Lag (ARDL) Modeling technique is utilized to find out the presence of long- and short-run relationship between trade openness and life expectancy. The findings indicated a stable long-run positive relationship between trade openness and life expectancy. Government expenditures, number of physicians and human capital growth have also positively and significantly impacted life expectancy. Similarly, growth of employment and number of beds in hospitals have influenced life expectancy negatively. Moreover, in the short run, government expenditures, growth of employment, number of physicians and human capital growth have maintained their relationship with life expectancy both in terms of coefficient signs and significance level while the relationship between trade openness, number of hospital beds and life expectancy is reversed.

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Trade openness; life expectancy; China; ARDL modeling

1. Introduction

The role of trade openness in the growth and development process of countries has been researched and debated indeed extensively over the last few decades. Trade openness is believed to be exerting positive impacts on the growth process of countries. Some of the prominent researchers such as Dollar (1992) and Frankel and Romer (1999) provided solid evidence in favor of significant positive relationship between trade openness and economic growth even after controlling the endogeneity problem. On the other side of the literature, Rodriguez and Rodrik (2000) and Rodríguez (2007) have identified several measurement and methodological problems in earlier literature and hence have put question mark regarding the positive trade-growth relationship. Moreover, the recent literature has again provided evidence in the support of the positive relationship that exists between trade openness and economic growth after addressing the contending issues (Tahir and Khan 2014; Tahir and Azid 2015). Fiestas (2005) commented that trade openness is not detrimental for economic growth despite some methodological problems.

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The literature on the trade-growth relationship is indeed very rich comprehensive. However, the role of trade openness in solving other main problems of developing countries is yet to be researched. Palamuleni (2017) pointed towards this issue and commented that only a few studies have focused on the relationship between trade openness and population health including life expectancy. Cowling et al. (2018) reviewed the literature and documented that only a few studies have explored the impacts of trade and investment on non-communicable diseases and risk characteristics. In this research paper, our primary objective is to highlight the contending role of trade openness in improving life expectancy. We have focused on the Chinese economy as the country has shown remarkable improvements both in life expectancy and trade openness over the last few decades. Zhang et al. (2018) documented that Chinese economy is constantly adopting and implementing open trade policies since joining World Trade Organization and further tariffs on imports are relatively low as compared to other countries. The current paper, therefore, tries to explore whether trade openness is a relevant factor in explaining the improved life expectancy enjoying by the huge population of China. As there are no studies on the linkages between trade and life expectancy in context of Chinese economy, results of this study would be therefore very useful for policymakers of China. They would be in turn in a much better position to devise appropriate policies both regarding life expectancy and trade openness.

Trade openness can affect health of the population through various ways. Firstly, trade openness brings significant drop in prices owing to the extensive competition in the host country and hence the real income increases particularly of the poor segment of the society. Consequently, increase in real income owing to trade openness enables the masses at large to get access to life-saving medicines and good quality consumable products and hence an overall positive impact on the quality of health would be seen. In a recent seminal paper, Jetter, Laudage, and Stadelmann (2019) showed a systematic and notable relationship between income and life expectancy by utilizing data of 197 countries for 213 years. Deaton (2004) argued that poverty is inversely related with life expectancy and hence it implies that if trade promotes poverty reduction, then it is logical to think that trade openness may influence the life expectancy positively. Herzer (2017) carried out a comprehensive study on the trade-health relationship and identified various channels including, income, inequality, access and aid channels by which trade openness can affect health of the population.

Herzer (2015) used historical data of the U.S. and provided evidence about the positive impact of trade openness on life expectancy. Timothy (2018) focused on the Nigerian economy and demonstrated that trade openness promotes life expectancy and suggested to policymakers that economic globalization shall be encouraged to improve life expectancy. Increased life expectancy could be useful as experience and knowledge which are the key elements of a modern growing economy associated with the increased life expectancy are of the view that trade openness influences health sector through income channel and also through direct distribution and provision of health-related goods and knowledge. Jawadi et al. (2018) focused on the member countries of the MENA region and found that trade openness has impacted life expectancy positively and infant mortality negatively. On the other hand, in the context of low-income economies, Palamuleni (2017) demonstrated a positive but bidirectional linkage between trade and life expectancy. Further, the author suggested the implementation of policies that

promote both trade openness and life expectancy. Herzer (2017) utilized data for five decades (1960–2010) for 74 countries and demonstrated a robust positive relationship between trade openness and life expectancy in the long run. Cyrus (2018) pointed out that trade openness is capable to enhance GDP due to which health outcomes would be improved in the long run. Moreover, Alam et al. (2016) also established a long-run relationship positive relationship between trade openness and life expectancy by focusing on Pakistan economy. It implies that the benefits of increased trade openness would be translated to the health sector and hence life expectancy would be flourished in the long run. It implies that the potential trade-health relationship would be of significant interest for policymakers.

This paper is structured as following. Section 2 describes basic statistics on both trade openness and life expectancy historically in China. Econometric methodology is discussed in Section 3. Results are analyzed and discussed in the penultimate section. The whole paper is concluded in the last section.

2. Descriptive statistics

This section is primarily devoted to discussing the historical evolution of both trade openness and life expectancy in China. Statistics are reported for the start and end year of the time chosen for the study. The purpose of this exercise is to provide comprehensive understanding about the behavior and trends of trade liberalization and life expectancy. Statistics are provided in Table 1.

The statistics given in Table 1 indicated that China has shown extraordinary growth in trade openness between 1970 and 2015. Trade openness which was hardly 4.952% of the GDP in 1970 has increased to 39.629% in 2015. The net increase in trade openness is observed to be more than 700%. The remarkable increase in trade openness reported in Table 1 reflects that the reforms of Chinese government over the years were successful. Lardy (2005) rightly commented that reforms of the government have transformed the Chinese economy from protectionism towards the most open economy. The observed increase in trade openness is indeed remarkable and this could be one of the main reasons of the notable economic growth of the Chinese economy during the last few decades.

In terms of life expectancy, the data presented in Table 1 indicated that China has done well to increase the life expectancy of its people. Life expectancy which was below 60 years on average in 1970 has increased to 75.928 years on average in 2015. The last column of Table 1 shows that life expectancy has been increased by 28.506% on average during the study period. Increase in life expectancy is a long-term phenomenon and hence a double digit increase during the last few decades is indeed commendable.

In the next step, we have shown the trend of both trade openness and life expectancy graphically for the period 1970–2015. It is inferred from Figure 1 that trade openness is

Table 1. Trade and life expectancy in China (1970–2015).

Variables	1970	2015	Percentage change
Trade openness	4.952	39.629	700.262
Life expectancy	59.085	75.928	28.506

Authors own calculation from WDI data.

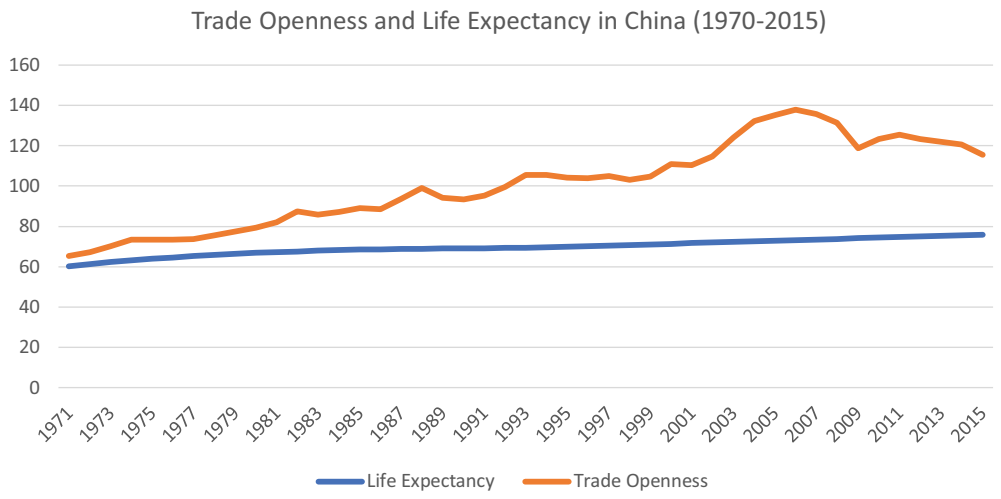

Figure 1. CUSUM test.

improved over the years steadily with slight variations until 2000. Upon acceptance to the WTO in 2001, trade openness of Chinese economy has increased significantly. The trade to GDP ratio has increased to 64.478% in 2006 which was highest in Chinese history. Between 2001 and 2006, trade openness of Chinese economy has been increased by 67.357% which is indeed a remarkable improvement. However, a downward trend is observed in trade openness during 2006 and 2009. Recent statistics shows that trade openness has recovered slightly for the Chinese economy and is improving steadily which is a good sign from the perspective of economic growth.

On the other hand, life expectancy has been increased steadily in China. Figure 1 demonstrates that life expectancy has shown an increasing trend during the study period. Both trade openness and life expectancy are having similar trend according to Figure 1 and hence a long-run relationship between the two is expected. Further, a correlation of 0.891 is observed between openness to trade and life expectancy which shows the strength of relationship between the two.

The observed increase both in terms of trade openness and longevity of life has motivated us to carry out this academic exercise for the Chinese economy. Therefore, results of this study would be indeed useful for policymakers of the Chinese economy. If trade openness matters for improved life expectancy, then policymakers would be in a much better position to implement appropriate policies for improving life expectancy using the policy of increased trade openness.

3. Modeling and methodology

3.1. The model

To explore the relationship of openness to trade with life expectancy, in this section, we specify the potential model. Life expectancy depends on several other factors besides trade openness. The literature has identified various factors responsible for the variations in life expectancy. For example, Reynolds and Avendano (2018) highlighted the role of

government expenditures in increasing life expectancy. Similarly, growth in employment level also affects the life expectancy through its impact on income level of the labor force. The studies of Delavari et al. (2016) and Luy et al. (2019) show that life expectancy is also dependent on the number of physicians and human capital. Further, the number of beds can also affect the life expectancy by improving the overall health conditions of the population. Therefore, we have used government expenditures, growth in employment, growth in human capital, number of physicians and number of hospital beds as control variables. The following model is specified in order to investigate the linkages between openness to trade and life expectancy for China.

$$\begin{aligned} lnlife_t = \ & \beta_0 + \beta_1 lntrade_t + \beta_2 gemp_t + \beta_3 lngex_t \\ & + \beta_4 lndoc_t + \beta_5 lnhos_t + \beta_6 ghc_t + U_t \end{aligned} \tag{1}$$

In Equation (1) life expectancy is the dependent variable and it is measured by the total number of years. Trade openness which is the variable of interest is captured through the ratio of exports plus imports to GDP. Price level of government consumption is used to approximate government expenditures. For the growth of employment, we have used the number of people engaged in millions. Number of physician and number of beds are measured per 1000 of population. Growth of human capital is measured by an index which is based on 'years of schooling and returns to education'. Data on life expectancy, number of physicians and hospital beds and trade openness are collected from WDI (World Development Indicators) while observations on government expenditures, human capital and employment level of labor force are obtained from PWT (Penn World Tables).

3.2. Estimating methodology

Time series data normally suffer from the unit root problem. The ordinary least square (OLS, hereafter) could not be utilized in circumstances where the unit root problem is associated with the variables, because unit root problem violates one of the dominant assumptions of OLS. Using OLS method of estimation in the presence of unit root problem would provide spurious results which would be misleading for policymakers.

Due to the mentioned problems, researchers have proposed some useful contending cointegrating approaches in order to handle time series data models. Cointegration in general means the stationary relationship that may exist between non-stationary variables. There are different cointegration tools available for handling time series data. The Engle and Granger (1987) approach is proposed in the literature which is suitable to handle two non-stationary variables. Similarly, Johansen (1988) developed a co-integration tool for handling multiple non-stationary variables. The Engle and Granger (1987) and Johansen (1988) approaches require variables to be non-stationary at level and stationary at first difference. However, it is possible that in some cases, some of the variables may be stationary and some may be non-stationary. Both the mentioned cointegration approaches are unable to handle variables of different order of cointegration. Pesaran, Shin, and Smith (2001) came up with a new cointegration approach, namely, Autoregressive Distributed Lag (ARDL, hereafter) model which works well

even if variables have similar or different order of integration. The ARDL approach is also suitable when the time dimension is short. Looking into the benefits, we therefore adopted the ARDL cointegrating approach in order to estimate model 1.

3.3. The ARDL modeling

The ARDL approach of cointegration was proposed recently by Pesaran, Shin, and Smith (2001). The ARDL modeling is suitable when the integration order is mixed. It can handle both stationary as well as non-stationary variables. Therefore, pre-testing of unit root is not important while employing the ARDL approach. Moreover, the ARDL is also appropriate for case studies where the time dimension is relatively small. Iheanacho (2017) pointed out that ARDL works well for the small size and provides unbiased results even if regressors are endogenous. Equation 1 is expressed in ARDL framework as shown below.

$$\begin{aligned}\Delta lnlife_t = \beta_0 &+ \sum\nolimits_{(i=1)}^{n1} \beta_{1i}\Delta lnlife_{(t-i)} + \sum\nolimits_{(i=0)}^{n2} \beta_{2i}\Delta lntrade_{(t-i)} \\ &+ \sum\nolimits_{(i=0)}^{n3} \beta_{3i}\Delta gemp_{(t-i)} + \sum\nolimits_{(i=0)}^{n4} \beta_{4i}\Delta lngex_{(t-i)} \\ &+ \sum\nolimits_{(i=0)}^{n5} \beta_{5i}\Delta lndoc_{t-i} + \sum\nolimits_{(i=0)}^{n6} \beta_{6i}\Delta lnhos_{t-i} \\ &+ \sum\nolimits_{(i=0)}^{n7} \beta_{7i}\Delta ghc_{t-i} + \gamma_0 lnlife_{t-1} + \gamma_1 lntrade_{t-1} \\ &+ \gamma_2 gemp_{t-1} + \gamma_3 lngex_{t-1} + \gamma_4 lndoc_{t-1} \\ &+ \gamma_5 lnhos_{t-1} + \gamma_6 ghc_{t-1} + \varepsilon_t\end{aligned} \tag{2}$$

$$\begin{aligned}\Delta nlntrade_t = \beta_0 &+ \sum\nolimits_{i=1}^{n1} \beta_1\Delta lntrade_{t-i} + \sum\nolimits_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} \\ &+ \sum\nolimits_{i=0}^{n3} \beta_{3i}\Delta gemp_{t-i} + \sum\nolimits_{i=0}^{n4} \beta_{4i}\Delta lngex_{t-i} + \sum\nolimits_{i=0}^{n5} \beta_{5i}\Delta lndoc_{t-i} \\ &+ \sum\nolimits_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} + \sum\nolimits_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \gamma_0 lnlife_{t-1} \\ &+ \gamma_1 lntrade_{t-1} + \gamma_2 gemp_{t-1} + \gamma_3 lngex_{t-1} + \gamma_4 lndoc_{t-1} \\ &+ \gamma_5 lnhos_{t-1} + \gamma_6 ghc_{t-1} + \varepsilon_t\end{aligned} \tag{3}$$

$$\begin{aligned}\Delta ngemp_t = \beta_0 &+ \sum\nolimits_{i=1}^{n1} \beta_{1i}\Delta gemp_{t-i} \\ &+ \sum\nolimits_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} \\ &+ \sum\nolimits_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} + \sum\nolimits_{i=0}^{n4} \beta_{4i}\Delta lngex_{t-i} \\ &+ \sum\nolimits_{i=0}^{n5} \beta_{5i}\Delta lndoc_{t-i} + \sum\nolimits_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} \\ &+ \sum\nolimits_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \gamma_0 lnlife_{t-1} \\ &+ \gamma_1 lntrade_{t-1} + \gamma_2 gemp_{t-1} \\ &+ \gamma_3 lngex_{t-1} + \gamma_4 lndoc_{t-1} + \gamma_5 lnhos_{t-1} + \gamma_6 ghc_{t-1} + \varepsilon_t\end{aligned} \tag{4}$$

$$\begin{aligned}\Delta nlngex_t &= \beta_0 + \sum_{i=1}^{n1} \beta_{1i}\Delta lngex_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} \\ &+ \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} + \sum_{i=0}^{n4} \beta_{4i}\Delta gemp_{t-i} \\ &+ \sum_{i=0}^{n5} \beta_{5i}\Delta lndoc_{t-i} + \sum_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} \\ &+ \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \gamma_0 lnlife_{t-1} + \gamma_1 lntrade_{t-1} + \gamma_2 gemp_{t-1} \\ &+ \gamma_3 lngex_{t-1} + \gamma_4 lndoc_{t-1} + \gamma_5 lnhos_{t-1} + \gamma_6 ghc_{t-1} + \varepsilon_t\end{aligned} \tag{5}$$

$$\begin{aligned}\Delta nlndoc_t &= \beta_0 + \sum_{i=1}^{n1} \beta_{1i}\Delta lndoc_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} \\ &+ \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} + \sum_{i=0}^{n4} \beta_{4i}\Delta gemp_{t-i} \\ &+ \sum_{i=0}^{n5} \beta_{5i}\Delta lngex_{t-i} + \sum_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} \\ &+ \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \gamma_0 lnlife_{t-1} + \gamma_1 lntrade_{t-1} + \gamma_2 gemp_{t-1} \\ &+ \gamma_3 lngex_{t-1} + \gamma_4 lndoc_{t-1} + \gamma_5 lnhos_{t-1} + \gamma_6 ghc_{t-1} + \varepsilon_t\end{aligned} \tag{6}$$

$$\begin{aligned}\Delta nlnhos_t &= \beta_0 + \sum_{i=1}^{n1} \beta_{1i}\Delta lnhos_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} \\ &+ \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} + \sum_{i=0}^{n4} \beta_{4i}\Delta gemp_{t-i} \\ &+ \sum_{i=0}^{n5} \beta_{5i}\Delta lngex_{t-i} + \sum_{i=0}^{n6} \beta_{6i}\Delta lndoc_{t-i} \\ &+ \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \gamma_0 lnlife_{t-1} + \gamma_1 lntrade_{t-1} \\ &+ \gamma_2 gemp_{t-1} + \gamma_3 lngex_{t-1} + \gamma_4 lndoc_{t-1} + \gamma_5 lnhos_{t-1} + \gamma_6 ghc_{t-1} + \varepsilon_t\end{aligned} \tag{7}$$

$$\begin{aligned}\Delta nghc_t &= \beta_0 + \sum_{i=1}^{n1} \beta_{1i}\Delta ghc_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} \\ &+ \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} + \sum_{i=0}^{n4} \beta_{4i}\Delta gemp_{t-i} + \sum_{i=0}^{n5} \beta_{5i}\Delta lngex_{t-i} \\ &+ \sum_{i=0}^{n6} \beta_{6i}\Delta lndoc_{t-i} + \sum_{i=0}^{n7} \beta_{7i}\Delta hos_{t-i} + \gamma_0 lnlife_{t-1} + \gamma_1 lntrade_{t-1} \\ &+ \gamma_2 gemp_{t-1} + \gamma_3 lngex_{t-1} + \gamma_4 lndoc_{t-1} + \gamma_5 lnhos_{t-1} + \gamma_6 ghc_{t-1} + \varepsilon_t\end{aligned} \tag{8}$$

In Equations (2)–(8), the parameters attached with the difference operators are capturing the short-run relationship. Similarly, the parameters associated with the lagged terms are designed to measure the long-run relationship between the dependent and independent variables. In the ARDL framework, the presence of long-run cointegration will be analyzed through the following null and alternative hypothesis.

$$H_0 : \gamma_0 = \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = \gamma_5 = \gamma_6 = 0 \tag{a}$$

$$H_1 : \gamma_0 \neq \gamma_1 \neq \gamma_2 \neq \gamma_3 \neq \gamma_4 \neq \gamma_5 \neq \gamma_6 \neq 0 \tag{b}$$

The null hypothesis assumes the absence while the alternative hypothesis accepts the presence of long-run cointegration relationship. Therefore, the rejection of null hypothesis would reflect the presence of long-run cointegration relationship and vice versa. The calculated F-test value would be compared with the tabulated values proposed by

Narayan (2004). There are three possible outcomes in the ARDL framework related to the presence or absence of long-run cointegration relationship. The presence of cointegration will be accepted in the scenario where the calculated F-statistic is greater than the upper bound critical value. Similarly, the absence of cointegration would be accepted if the calculated F-statistic falls below the lower bound critical value. Lastly, the presence of cointegration would be inconclusive if the calculated F-statistic lies between the lower and upper bound critical values.

4. Results and discussion

4.1. Unit root testing

In the first instance, we have carried out the unit root testing using the well-known Augmented Dickey Fuller Test (ADF, hereafter). Although the ARDL approach the pre-testing of unit root is not desirable, however, still it is necessary to identify the order of integration and make sure that second-differenced stationary variables are not present as the ARDL approach does not handle variables if their order of integration is more than one.

Results for the stationarity testing based on ADF test are depicted in Table 2. According to results, all variables except human capital growth are non-stationary at level. Human capital growth is stationary at level. However, the results presented in the third column of Table 2 reflected that the unit root problem disappeared for all variables while taking the first difference. Hence, it could be concluded that the selected variables are stationary on the first difference. The mixed order of integration of variables is one of the motivations behind using ARDL approach.

4.2. Bound testing results

After identifying the order of integration, we conducted the bound testing in order to examine the presence of long-run cointegrating relationship. Results are presented in Table 3.

Results for the bounding testing which are based on the F-test are shown in Table 3. Results provided concrete evidence in the support of strong cointegrating relationship among the variables. According to results, the null hypothesis of no cointegration is not accepted in all cases. The calculated F-test values are greater than the upper bound critical values. Therefore, we conclude based on the evidence presented in Table 3 that the long-run cointegration relationship is present among the chosen variables.

Table 2. Unit root testing.

Variables	Level	First difference	Decision
$lnlife_t$	−2.746	−4.115***	I(1)
$lntrade_t$	−0.896	−4.737***	I(1)
$gemp_t$	−2.900	−6.491***	I(1)
$lngex_t$	−1.096	−4.223***	I(1)
$lndoc_t$	−2.483	−8.898***	I(1)
$lnhos_t$	−2.225	−8.330***	I(1)
ghc_t	−7.192***	−5.160***	I(0)

Author own calculation using Eviews-9.

Table 3. The bound testing results.

Dependent variables	*F*-test	Decision
F($lnlife_t$/$lntrade_t$,$gemp_t$,$lngex_t$,ghc_t,$lndoc_t$,$lnhos_t$)	245.341	Co-integrated
F($lntrade_t$/$lnlife_t$,$lnemp_t$, $lngex_t$, ghc_t,$lndoc_t$,$lnhos_t$)	3.604	Co-integrated
F ($gemp_t$/$lnlife_t$,$lntrade_t$, $lngex_t$, ghc_t,$lndoc_t$,$lnhos_t$)	5.237	Co-integrated
F ($lngex_t$/$lnlife_t$,$lntrade_t$, $gemp_t$,ghc_t,$lndoc_t$,$lnhos_t$)	5.161	Co-integrated
F (ghc_t/$lnlife_t$,$lntrade_t$, $gemp_t$,$lngex_t$,$lndoc_t$,$lnhos_t$)	7.613	Co-integrated
F ($lndoc_t$/$lnlife_t$,$lntrade_t$, $gemp_t$,$lngex_t$,ghc_t,$lnhos_t$)	5.852	Co-integrated
F ($lnhos_t$/$lnlife_t$,$lntrade_t$, $gemp_t$,$lngex_t$,ghc_t,$lndoc_t$)	7.011	Co-integrated
Critical values	Lower Bound I(0)	Upper Bound I (1)
1%	3.15	4.43
5%	2.45	3.61
10%	2.12	3.23

Note: (***) represents 1 % significance level. Authors own calculation from E-views 9.0.

In the next step, we specify the error correction models. The purpose of the error correction modeling is to show the stability of the long-run coefficients and further examine the speed of adjustment of the model. The error correction models are specified in the following manner.

$$\begin{aligned}\Delta nlnlife_t = \beta_0 &+ \sum_{i=1}^{n1} \beta_{1i}\Delta lnlife_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lntrade_{t-i} \\ &+ \sum_{i=0}^{n3} \beta_{3i}\Delta gemp_{t-i} + \sum_{i=0}^{n4} \beta_{4i}\Delta lngex_{t-i} + \sum_{i=0}^{n5} \beta_{5i}\Delta lndoc_{t-i} \\ &+ \sum_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} + \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \theta_1 ECT_{t-1} + \varepsilon_t\end{aligned} \tag{9}$$

$$\begin{aligned}\Delta nlntrade_t = \beta_0 &+ \sum_{i=1}^{n1} \beta_{1i}\Delta lntrade_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} + \sum_{i=0}^{n3} \beta_{3i}\Delta gemp_{t-i} \\ &+ \sum_{i=0}^{n4} \beta_{4i}\Delta lngex_{t-i} + \sum_{i=0}^{n5} \beta_{5i}\Delta lndoc_{t-i} + \sum_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} \\ &+ \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \theta_2 ECT_{t-1} + \varepsilon_t\end{aligned} \tag{10}$$

$$\begin{aligned}\Delta ngemp_t = \beta_0 &+ \sum_{i=1}^{n1} \beta_{1i}\Delta gemp_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} + \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} \\ &+ \sum_{i=0}^{n4} \beta_{4i}\Delta lngex_{t-i} + \sum_{i=0}^{n5} \beta_{5i}\Delta lndoc_{t-i} \\ &+ \sum_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} + \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \theta_3 ECT_{t-1} + \varepsilon_t\end{aligned} \tag{11}$$

$$\begin{aligned}\Delta nlngex_t = \beta_0 &+ \sum_{i=1}^{n1} \beta_{1i}\Delta lngex_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} + \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} \\ &+ \sum_{i=0}^{n4} \beta_{4i}\Delta gemp_{t-i} + \sum_{i=0}^{n5} \beta_{5i}\Delta lndoc_{t-i} + \sum_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} \\ &+ \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \theta_4 ECT_{t-1} + \varepsilon_t\end{aligned} \tag{12}$$

$$\Delta nlndoc_t = \beta_0 + \sum_{i=1}^{n1} \beta_{1i}\Delta lndoc_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} + \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} + \sum_{i=0}^{n4} \beta_{4i}\Delta gemp_{t-i} + \sum_{i=0}^{n5} \beta_{5i}\Delta lngex_{t-i} + \sum_{i=0}^{n6} \beta_{6i}\Delta lnhos_{t-i} + \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \theta_5 ECT_{t-1} + \varepsilon_t \tag{13}$$

$$\Delta nlnhos_t = \beta_0 + \sum_{i=1}^{n1} \beta_{1i}\Delta lnhos_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} + \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} + \sum_{i=0}^{n4} \beta_{4i}\Delta gemp_{t-i} + \sum_{i=0}^{n5} \beta_{5i}\Delta lngex_{t-i} + \sum_{i=0}^{n6} \beta_{6i}\Delta lndoc_{t-i} + \sum_{i=0}^{n7} \beta_{7i}\Delta ghc_{t-i} + \theta_6 ECT_{t-1} + \varepsilon_t \tag{14}$$

$$\Delta nghc_t = \beta_0 + \sum_{i=1}^{n1} \beta_{1i}\Delta ghc_{t-i} + \sum_{i=0}^{n2} \beta_{2i}\Delta lnlife_{t-i} + \sum_{i=0}^{n3} \beta_{3i}\Delta lntrade_{t-i} + \sum_{i=0}^{n4} \beta_{4i}\Delta gemp_{t-i} + \sum_{i=0}^{n5} \beta_{5i}\Delta lngex_{t-i} + \sum_{i=0}^{n6} \beta_{6i}\Delta lndoc_{t-i} + \sum_{i=0}^{n7} \beta_{7i}\Delta hos_{t-i} + \theta_7 ECT_{t-1} + \varepsilon_t \tag{15}$$

Equations (9)–(15) are the error correction models where the parameters attached with the difference operator are measuring the short-run dynamics. The error correction terms included in Equations (9)–(15) basically capture the speed of adjustment. Ideally, it should be negative and significant. Results of short-run relationship based on error correction model are presented in the bottom part of Table 4.

4.3. Discussion on long-run and short-run findings

After establishing the presence of long-run cointegration relationship, in this section have presented both the long-run and short-run coefficients in Table 4.

Table 4. Long-run and short-run relationship.

Variables	Coefficients	Standard Errors	*T*-test
Long Run			
$Intrade_t$	0.0711***	0.0092	7.7023
$lngex_t$	0.0600***	0.0058	10.2185
$gemp_t$	−0.3533***	0.1061	−3.3292
ghc_t	4.8131***	1.1385	4.2275
$lndoc_t$	0.0254*	0.0146	1.7374
$lnhos_t$	−0.0399***	0.0086	−4.5990
Short Run			
$\Delta lntrade_t$	−0.8803***	0.0131	−66.8467
$\Delta lngex_t$	0.0003***	0.0001	3.0000
$\Delta gemp_t$	−0.0040***	0.0012	−3.1686
Δghc_t	0.0006	0.0002	3.0000
$\Delta lndoc_t$	0.0049***	0.0015	3.3022
$\Delta lnhos_t$	0.0001	0.0002	0.5000
ECT (−1)	−0.0114***	0.0020	−5.4964
Adj R^2: 0.999			

The asterisk (***) (**) and (*) stands for 1, 5 and 10 % percent significance level, respectively.

Both long- and short-run results are highlighted in Table 4. In the upper part of Table 4, the long-run results are presented between the dependent and independent variables. It is inferred from the presented results that the impact of trade openness on life expectancy is not only positive but also different from zero statistically. To put it differently, trade openness has contributed significantly to life expectancy of the Chinese people over the years. The positive impact of trade openness on life expectancy is also consistent with Herzer (2017) who also reported a positive and robust impact of trade openness on life expectancy by focusing on the developing countries. Therefore, policymakers of China should further liberalize international trade in order to not only improve economic growth but also to enhance the life expectancy of the masses.

Similarly, the results also showed that government expenditures have also influenced life expectancy both positively and significantly. It means that the Chinese government has played a significant and vital role in uplifting the life expectancy of people by incurring huge expenditures. Government expenditures especially in developing countries like China are important for the welfare of the masses. Kofi et al. (2018) also documented that public health expenditures matters for health outcomes. Policymakers are therefore required to enhance government expenditures to further improve life expectancy. It is an undeniable fact that old and experienced people are in a much better position to contribute to the growth and development of countries. Therefore, the Chinese government can take advantage of the experience and knowledge of their people by improving their life expectancy through increased government expenditures.

We also found that growth of employment is adversely connected to life expectancy which is against our prior expectation. The possible reasons behind the observed negative relationship between growth of employment and life expectancy could be because of the fact that employed people are mostly suffering from various problems such as stress and tensions due to which their life cycle shortens.

Moreover, the results indicated that the growth of human capital over the years has influenced the life expectancy both positively and significantly. Human capital level of the population can influence life expectancy as it creates awareness about the quality of life among the general population. The observed positive impact of human capital growth on life expectancy is similar with the findings of Luy et al. (2019) who commented that human capital policies can be recognized as indirect health policies. Further, Kaplan, Spittel, and Zeno (2014) also reviewed the literature and reported that human capital is one of the best predictors of life expectancy.

Moreover, it is observed that the number of physicians has played a commendable role in improving the life expectancy. Number of physicians can enhance the life expectancy by improving the overall health conditions of the population of the masses by curing deadly diseases through advanced medical treatment. Increase in the number of physicians gives access to medical services due to which the utilization of health services improves and consequently life expectancy enhances (Delavari et al. 2016). On the contrary, the findings showed that the number of hospital beds have adversely influenced life expectancy. One of the plausible reasons behind the negative association could be the building of new hospitals without having special focuses on deadly diseases. Medical resources deployed in the generalized hospitals could have positively contributed in the life expectancy if they were utilized in providing specialized treatment to the patients. That is why the increased number of hospital beds in the generalized hospitals could not

result in higher life expectancy. The negative relationship between number of beds and life expectancy is also consistent with the findings of Jebeli, Hadian, and Souresrafil (2019). The number of physicians appeared to be more than important than the number of beds from the perspective of improved life expectancy. Therefore, Chinese health authorities are suggested to increase the number of physicians to increase life expectancy of the population further.

In short run, we found that the positive impact of trade openness on life expectancy disappeared. According to the results, trade openness has an inverse relationship with life expectancy in short run, unlike the positive relationship which was observed in the long run. In the short run, when the domestic economy opens towards the global economy, workers in the import competing industries suffer and hence their incomes decrease drastically. Consequently, their standard of living falls due to which life expectancy decreases. Similarly, it is also possible that in the short-run trade openness may encourage the consumption of processing foods and other harmful goods owing to opening markets and lower prices and consequently life expectancy may be worsened (Cyrus 2018). It means that in short run, trade openness may cast adverse impact on life expectancy. Moreover, in a recent article, Scott and Blessing (2019) empirically demonstrated a negative relationship between trade openness and life expectancy for the Nigerian economy in the short run. They argued that in the short run, the inclusion of processed consumer goods in imports may be harmful for improved life expectancy. A strict monitoring over processed foods and other harmful products, therefore, is required in order to ensure safety of the general population in the short run. The observed negative relationship between trade openness and life expectancy in the short run implies that trade openness may not be an immediate solution to increase life expectancy. In other words, the benefits of trade openness for improved life expectancy would be materialized only in the long run. However, sustained trade openness over a long period of time would influence life expectancy positively and significantly. Therefore, policymakers of the Chinese economy are suggested to continue the trade liberalization policies. These policies will not only boost economic growth but would also increase life expectancy substantially.

Further, in the short run, we found that government expenditures have impacted life expectancy both positively and significantly. The positive linkages of government expenditures with life expectancy in short and long run are a clear indication that government role is important for improving the life expectancy of population. Government programs are mostly having long-lasting impacts and hence shall be continued in order to improve the life expectancy of upcoming generations. Lastly, in the short run, the results indicated that growth of employment has impacted life expectancy negatively and significantly like the earlier findings. The negative relationship between growth of employment in the short as well as in the long run reflect the fact that employed individuals are having shorter life span owing to multiple problems including stress and tension.

Finally, the positive relationship between growth of human capital, number of physicians and life expectancy observed in the long run did not alter in the short run. It implies that both should be given priority in order to improve life expectancy. Surprisingly, in the short run, the relationship between number of beds and life expectancy turned positive although it is insignificant statistically. It implies that in the short run basic medical facilities such as number of beds are required for the overall health improvement including life expectancy.

The error correction term entered to the estimated regression models both negatively and significantly. The negative coefficient sign and significance of the error correction terms is the indication of the stability of long-run findings. Further, coefficient of the error correction term which shows speed of adjustment from the short-run disequilibrium towards the long-run equilibrium is not very high. The low speed of adjustment of the model is the indication that trade openness and life expectancy have a long-run relationship.

4.4. Diagnostic checking

In this section, we have carried out some important diagnostic checking in order to confirm that the specified and estimated model is free from any serious econometric issue. Tests for serial correlation, normality, heteroscedasticity and functional from are conducted. Table 5 includes results of the aforementioned tests.

According to findings given in Table 5, the estimated model has no serial correlation as the null hypothesis of no serial correlation cannot be rejected at 5% significance level. Similarly, the ARCH test rejected the presence of heteroscedasticity as the null hypothesis cannot be rejected at standard level. Moreover, the Jarque-Bera and Ramsey tests have confirmed the normality of data and correct functional form of the model, respectively.

4.5. Residuals stability

In addition to the tests given in Table 5, we have also carried out CUSUM and Square of CUSUM tests in order to test the stability of residuals. Both the tests are graphed in Figure 1 and Figure 2. Both the tests have confirmed the stability of the residuals as they are lying within the critical boundaries. Therefore, we can conclude that the long- and short-run findings obtained are stable.

5. Conclusions and implications

5.1. Conclusions

This paper tried to explore the potential relationship between trade openness and life expectancy for the Chinese economy. Annual time series data for the period 1970–2015 are collected from the internationally reliable sources. The ARDL modeling approach is adopted for testing the presence of long-run relationship between trade openness and life expectancy. After the confirmation of long-run relationship between trade openness and life expectancy, in the next step, both short and long-run coefficients are calculated Figure 3.

Table 5. Diagnostic checking.

Diagnostics	Null hypothesis	*F*-test	Conclusion
LM Test (Serial Correlation	H_0: No Serial Correlation	2.577 (0.275)	Don't reject H_0
ARCH Test (Heteroscedasticity)	H_0: No Heteroscedasticity	2.077 (0.149)	Don't reject H_0
Jarque-Bera (Normality)	H_0: Residuals are Normally Distributed	1.115 (0.572)	Don't reject H_0
Ramsey Test (Functional Form)	H_0: Functional Form is Correct	0.834 (0.376)	Don't reject H_0

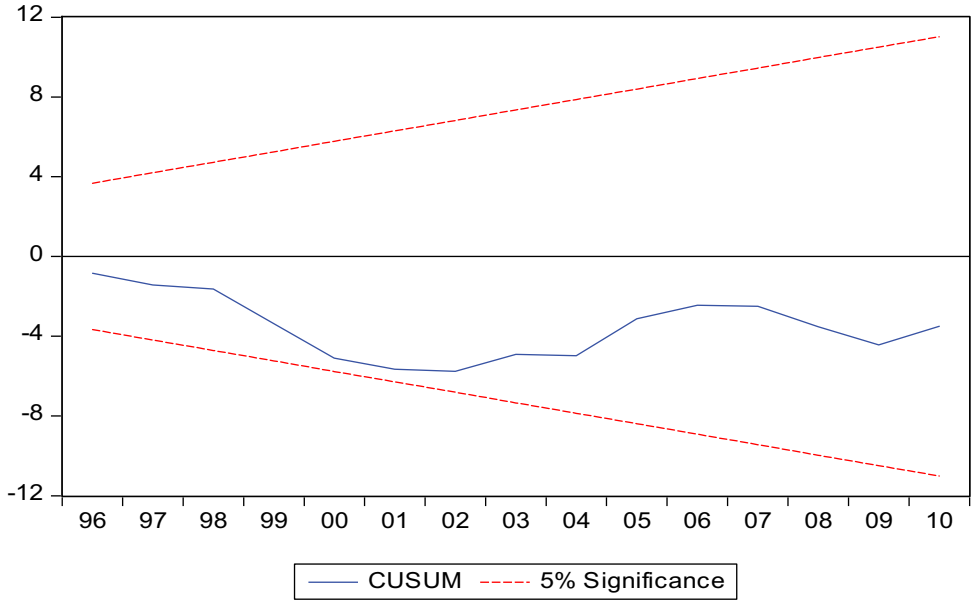

Figure 2. CUSUM Test.

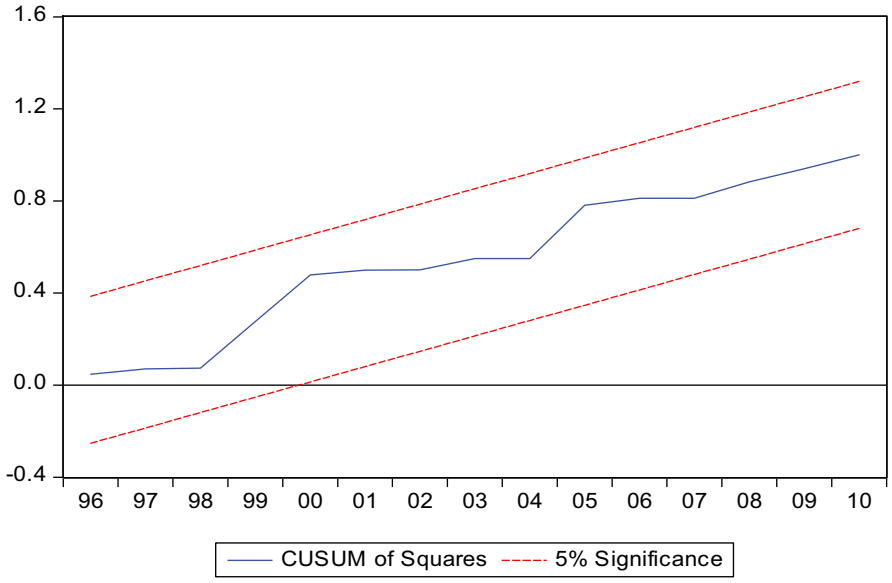

Figure 3. CUSUM of Squares

According to the long-run results, trade openness has accelerated life expectancy of the population of China not only positively but also significantly. It implies that higher trade openness matters for higher life expectancy which is desirable for the welfare point of view. Higher life expectancy also matters for the growth and development of

countries owing to the experience and knowledge of older people. Similarly, we also found evidence in the support of higher government expenditures, number of physicians, human capital growth and higher life expectancy. It implies that the Chinese government should increase their expenditures, increase number of physicians and invest in education in order to further enhance life expectancy of ordinary Chinese people. Moreover, we found that growth of employment is detrimental for life expectancy which is against our prior expectation. There could be many factors such as tensions and stress prevailing in workplace responsible for the observed negative impact of growth of employment on life expectancy. Finally, the results showed that the number of beds in hospital have not improved life expectancy in China. It could be because of the fact that the required number of beds are already available. The study suggests that Chinese government and policymakers should further liberalize trade, increase government expenditures, increases number of physicians, invest more in human capital and re-visit the working environment of Chinese-employed people in order to not only improve economic growth but also increase life expectancy which is desirable from the welfare point of view.

5.2. Recommendations

Based on the comprehensive analysis carried out in this paper, we recommend the following points to policymakers of China.

(1) Chinese policymakers are advised to further liberalize trade for improving not only growth but also life expectancy which is desirable from the welfare point of view.
(2) The Chinese government is suggested to increase its expenditures, increase number of physicians and further invest more in human capital as they are correlated with life expectancy. Government expenditures are long lasting and hence current expenditures would increase life expectancy in the coming decades.
(3) Chinese policy makers are also suggested to re-visit the working environment of the employed people and bring significant reforms. These reforms would ease both the physical and mental stress and hence their life expectancy would increase significantly.

Disclosure statement

No potential conflict of interest was reported by the author.

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